

Optical Fiber Distributed Acoustic Sensors: A Review

Zuyuan He , Senior Member, IEEE, Senior Member, OSA, and Qingwen Liu , Member, IEEE

(Invited Tutorial)

Abstract—Fiber-optic distributed acoustic sensor (DAS) is one of the most attractive and promising fiber-optic sensing technologies in the recent decade. It can simultaneously detect and retrieve multiple vibrations over a long distance, and the high sampling rate provides abundant information of the environment. This article reviews the principles involved in DAS system, including three types of reflectometry to locate the Rayleigh backscattering (RBS) along the fiber, and the methods to recover the vibration waveform by the phase or spectrum of RBS. The technologies and recent progresses on DAS systems are introduced, and two kinds of typical applications of DAS are reviewed. Finally, the possible research trends are discussed.

Index Terms—Distributed acoustic sensor, fiber sensor, rayleigh backscattering.

I. INTRODUCTION

THE optical fiber distributed acoustic sensor (DAS) utilizes the Rayleigh backscattering (RBS) to detect and locate the vibrations along the optical fiber [1]. It can obtain full information of vibration (amplitude, frequency and phase) at all positions along the optical fiber, whose length is up to 100 kilometers. Together with its advantages like small size, immune to electromagnetic interference, robust against harsh environment, the DAS is a promising solution in many applications, especially in the long distance applications like pipeline security, and harsh environments like oil and gas wells.

Compared with the well-developed distributed temperature sensor (DTS) based on Raman backscattering and the distributed strain / temperature sensor based on Brillouin backscattering, the DAS has its unique advantages. First, the vibration can propagate a long distance in medium, so the DAS can detect the event which is far away from the fiber cable. As a result, the coverage of DAS is large; Second, the RBS is the elastic spreading of the incident lightwave. The DAS is based on the interference of the RBS of the high coherence probe lightwave within a certain range (spatial resolution), so high measuring sensitivity can be achieved. As a comparison, the Brillouin backscattering can't keep the coherence of the pump lightwave. The highest strain

resolution is about several micro-strain [2], which is orders of magnitude lower than the RBS-based DAS. Third, the RBS is stronger than the spontaneous Brillouin scattering and Raman backscattering in fiber. Combined with the high sensitivity, the DAS can carry out one measurement by using one probe pulse without average. So the sampling rate can be from kHz to MHz, providing sufficient information of the event.

Early technologies based on the variation of RBS intensity can only detect the existence of vibration, and cannot reproduce the amplitude or phase correctly. It is usually called distributed vibration sensor (DVS) [3]. Later, technologies based on phase of RBS can accurately restore the amplitude, frequency and phase information of vibration, and it is named as distributed acoustic sensor (DAS). The high performance DAS can obtain the waveform of the vibration with high definition, so it is possible to extract many parameters of the measurand via proper data processing techniques such as array processing. For example, the phases of all sensing points are matched, so combining the acoustic response characteristics of the optical cable can increase the sensitivity of signal detection. With those characteristics the DAS technology has better performance than DVS, and can complete more complex tasks, such as measuring the ratio of oil and water in the pipeline by detecting the propagation speed of sound waves in the surrounding medium of the fiber.

This article reviews the principle and recent progress of the DAS. Generally, the DAS system has two main functions: One is to locate the RBS at a certain position on the fiber. Reflectometry is the technology to undertake this task. Currently, most DAS systems are based on optical time domain reflectometry (OTDR) scheme, which has simple structure, low cost, and easy demodulation algorithm. Other types of reflectometry are also adopted in DAS system for enhanced spatial resolution or sensitivity. Three types of reflectometry including OTDR will be introduced in Section II. The other function of DAS is to retrieve the vibration signal from the RBS. The vibration will change the interference state of the RBS, resulting in both phase shift and intensity fluctuation. Section III will discuss the principle of vibration signal recovery via phase demodulation, and the spectral demodulation technology will be introduced in Section IV. Section V gives some practical applications of DAS and signal processing technologies. A conclusion and a prospect of DAS are drawn at the last section.

II. REFLECTOMETRIES FOR DAS

In DAS system, there are two key parameters for the reflectometry. One is the spatial resolution, which indicates the ability

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The authors are with the State Key Laboratory of Advanced Optical Communication Systems and Networks, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: zuyuanhe@sjtu.edu.cn; liuqingwen@sjtu.edu.cn).

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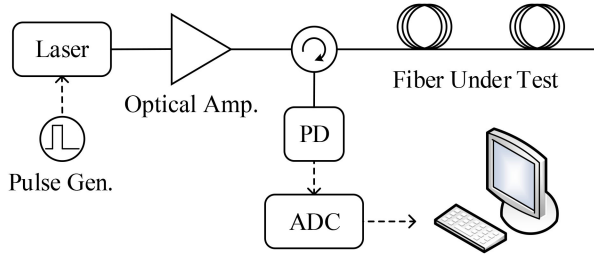


Fig. 1. A typical OTDR configuration. PD: photo-detector; ADC: Analog to Digital Converter.

to separate the RBS from different positions and thus determines the ability to distinguish nearby vibration events along the fiber. The other one is the signal-to-noise ratio (SNR) of the detected RBS signal. As will be discussed later, the sensitivity of vibration detection depends on the SNR. **Since the SNR decreases due to the attenuation of the fiber, the maximum sensing distance is determined by the minimum acceptable SNR and the attenuation coefficient of the fiber.** Distributed amplification technology like Raman amplification can be adopted to compensate the attenuation of the fiber, and thus achieves long sensing distance [4].

Although most DAS are based on the OTDR configuration, other types of reflectometries have also been investigated for vibration location. The optical frequency domain reflectometry (OFDR) is employed for locating RBS in [5]–[10]. **Compared with OTDR, OFDR's main advantages are higher spatial resolution and high signal-to-noise ratio (SNR). But it is very sensitive to laser phase noise, and the quasi-continuous probe boosts the cross-talk problem, so OFDR systems generally have shorter sensing distances with low response bandwidth.** Recently, a time-gated digital optical frequency domain reflectometry (TGD-OFDR) [11], [12] has been adopted in DAS system, showing the advantages of both OTDR and OFDR. In this section, the above three reflectometry technologies will be discussed in detail.

A. Optical Time Domain Reflectometry (OTDR)

To monitor the health of the optical communication link, OTDR is firstly proposed by Barnoski and Jensen in 1976 [13]. A typical OTDR configuration is shown in Fig. 1. A short optical pulse is injected into the fiber under test (FUT). The optical pulse propagates forward along the FUT, and RBS signals are generated at different positions with different round-trip time τ . Thus, the distance along the FUT is mapped to the time delay τ :

$$z = \frac{c\tau}{2n}, \quad (1)$$

where c is the velocity of lightwave in vacuum, and n is the refractive index of fiber.

To improve the SNR of the OTDR system, coherent detection is employed. A typical coherent OTDR (COTDR) configuration is illustrated in Fig. 2. The difference from traditional OTDR system is that the laser source is a narrow linewidth laser with a long coherence length. In a COTDR system, the electrical signal generated by the arbitrary waveform generator (AWG) is

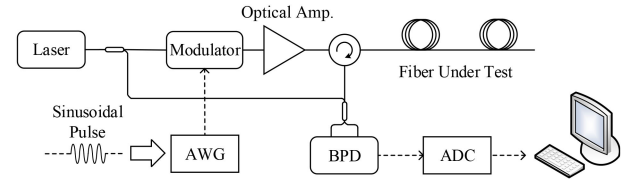


Fig. 2. COTDR configuration. AWG: arbitrary waveform generator; BPD: balanced photo-detector.

expressed as:

$$s(t) = w\left(\frac{t}{T_p}\right) e^{j2\pi f_0 t}, \quad (2)$$

where T_p is the probe pulse width, $w(t)$ is the window function, and f_0 is the carrier frequency (also is the frequency shift of the optical probe pulse).

The high coherence lightwave from the laser is split into two beams. One beam is modulated as the probe pulse, and the other one is the local oscillator (LO). Thus, the electrical fields before and after the modulation are expressed as:

$$E_l(t) = \sqrt{P_l} e^{j2\pi\nu_0 t}, \quad (3)$$

and

$$E_p(t) = \sqrt{P_p} w\left(\frac{t}{T_p}\right) e^{j(2\pi\nu_0 t + 2\pi f_0 t)}, \quad (4)$$

where P_p and P_l are the power of the probe light and LO respectively, and ν_0 is the center frequency of the laser. The photoelectrical field received by the balanced photo-detector (BPD) is the summation of all RBS signals along the FUT:

$$E_R(t) = \sum_{i=1}^N E_i w\left(\frac{t - \tau_i}{T_p}\right) e^{j[2\pi\nu_0(t - \tau_i) + 2\pi f_0(t - \tau_i)]}, \quad (5)$$

where N is the number of the scattering points along the FUT, τ_i is the round-trip time of the i th scattering point and E_i is its photoelectrical field amplitude, which is determined by the reflectivity of the scattering point and the attenuation of the fiber.

The photocurrent generated at BPD is:

$$\begin{aligned} i(t) &\propto \Re\{E_R(t) \cdot E_l^*(t)\} \\ &= \sum_{i=1}^N A_i w\left(\frac{t - \tau_i}{T_p}\right) \cos[2\pi f_0(t - \tau_i) - 2\pi\nu_0\tau_i], \end{aligned} \quad (6)$$

where $\Re\{E\}$ means taking the real part of the complex E , $*$ is the conjugation and A_i is the photocurrent amplitude converted from the RBS signal of the i th scattering point.

Hilbert transform can be employed to convert the real signal to analytic signal. Since N in Eq. (6) is large enough and the distance between two adjacent scattering points is smaller than the optical wavelength, Eq. (6) can be rewritten as an integral:

$$\begin{aligned} i(t) &= \int_0^{T_F} A(\tau) w\left(\frac{t - \tau}{T_p}\right) e^{j[2\pi f_0(t - \tau) - 2\pi\nu_0\tau]} d\tau \\ &= h(t) \otimes s(t), \end{aligned} \quad (7)$$

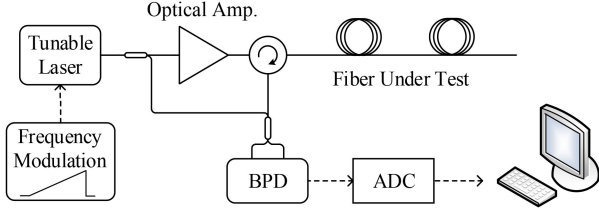


Fig. 3. OFDR configuration.

where $\otimes$ is the convolution operation, T_F is the maximum round-trip time along the FUT and $h(t)$ is defined as the impulse response of the FUT:

$$h(t) = A(t)e^{-j2\pi\nu_0 t}. \quad (8)$$

Therefore, COTDR system can be regarded as a linear response system. The input of the system $s(t)$ is the electrical signal generated by the AWG. The impulse response of the system is $h(t)$, which represents the phase and intensity change of the injecting light in the FUT. The output of the system is the photocurrent $i(t)$, from which environmental parameters (vibrations) can be demodulated.

The time resolution of $i(t)$ is the same as that of $s(t)$. So the spatial resolution of OTDR (and COTDR) is:

$$Z_{SR} = \frac{cT_p}{2n}. \quad (9)$$

The SNR of RBS is defined as the ratio between the RBS power and the noise floor. It is a function of pulse power, pulse width and sensitivity of photo-detector. Increasing the pulse power can improve the SNR, but it is ultimately limited to avoid the onset of the undesirable nonlinear effects [14]. Higher spatial resolution requires a narrower pulse width T_p , which will bring lower pulse energy, broader receiver bandwidth, and higher noise level, so the SNR will be poor. **The trade-off between the spatial resolution and the SNR is the main challenge faced by OTDR.** It is suitable for applications requiring long sensing range.

B. Optical Frequency Domain Reflectometry

Optical frequency domain reflectometry (OFDR) is widely employed to monitor the optical link with high spatial resolution [15]–[19]. **OFDR has high SNR because a continuous wave (CW) is employed as the probe, and the spatial resolution is determined by the frequency sweeping range.** What's more, the bandwidth of the receiver is much lower than the frequency tuning range of the probe light.

A typical OFDR configuration is illustrated in Fig. 3. The linearly frequency sweeping lightwave from the source is split into two beams. One beam is the LO and the other one is the probe light. The LO is expressed as:

$$E_l(t) = \sqrt{P_l}w\left(\frac{t}{T_s}\right)e^{j[2\pi\nu_0 t + \pi\gamma t^2 + \phi(t)]}, \quad (10)$$

where P_l is the power of the light, ν_0 is the center frequency, γ is the sweeping rate, $\phi(t)$ is the phase noise term, and T_s is the frequency sweeping duration. **The phase noise is caused by frequency sweeping nonlinearity and spontaneous radiation of the laser.** In OFDR system, the total round-trip time of the fiber

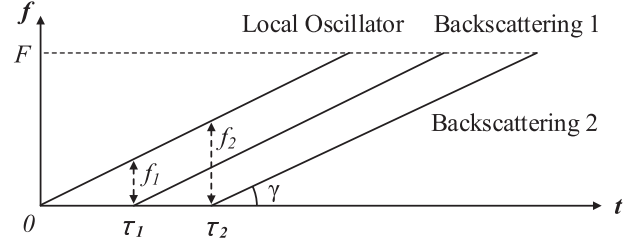


Fig. 4. Locating principle of OFDR. The time delay of RBS is mapped to the frequency of the beat signal.

T_F is much smaller than the frequency sweeping duration T_s , i.e., $T_F \ll T_s$.

Since the probe light and the LO are from the same light source, the RBS of the i th scattering point is the time-delayed copy of the LO, which is:

$$E_i^{RBS}(t) = E_i w\left(\frac{t - \tau_i}{T_s}\right) \exp\{j[2\pi\nu_0(t - \tau_i) + \pi\gamma(t - \tau_i)^2 + \phi(t - \tau_i)]\}, \quad (11)$$

where τ_i is the round-trip time of the i th scattering point and E_i is its photoelectrical field intensity. Since $\tau_i \ll T_s$, $w(\frac{t - \tau_i}{T_s}) \approx w(\frac{t}{T_s})$.

The received photocurrent signal is proportional to the summation of the beating signals of all scattering points:

$$i(t) \propto \sum_{i=1}^N \Re\{E_i^{RBS*}(t) \cdot E_l(t)\} = \sum_{i=1}^N A_i \cos[2\pi\gamma\tau_i t + 2\pi\nu_0\tau_i - \pi\gamma\tau_i^2 + \phi(t) - \phi(t - \tau_i)]w\left(\frac{t}{T_s}\right). \quad (12)$$

The RBS at different point has different round-trip time, which results in different beating frequencies with the LO, as shown in Fig. 4. Then Fourier transform (FT) is performed to separate the beating signals of different points.

When the sensing distance is much smaller than the coherence length and the center frequency of the light source sweeps linearly, the phase noise term $\phi(t) - \phi(t - \tau_i)$ can be ignored. Thus, the FT of the Eq. (12) is:

$$I(f) = \sum_{i=1}^N A_i \exp\{j[2\pi\nu_0\tau_i - \pi\gamma\tau_i^2]\}W(f - \gamma\tau_i) \quad (13)$$

where $W(f)$ is the Fourier transform of the $w(t)$. Here only the positive part of the FT is considered since the FT of a real signal is symmetrical. In Eq. (13), the beating frequency of the i th scattering point is $f_i = \gamma\tau_i$ and its phase is $\Phi_i = 2\pi\nu_0\tau_i - \pi\gamma\tau_i^2$.

According to Eq. (13), both the intensity and phase of the RBS along the FUT are obtained via FT [10]. Similar to Eq. (7), Eq. (13) can be rewritten as an integral:

$$I(f) = \int_0^{\gamma T_F} A(\nu) \exp\{j\Phi(\nu)\}W(f - \nu)d\nu = H(f) \otimes W(f), \quad (14)$$

where T_F is the maximum round trip time of the FUT and $H(f)$ is the impulse response of FUT in frequency domain:

$$H(f) = A(f) \exp\{j[2\pi\nu_0 f/\gamma - \pi f^2/\gamma]\}. \quad (15)$$

Therefore, the OFDR system is a linear response system in frequency domain. So it is also a linear response system in time domain. The FT of the time window $W(f)$ is the input of the system. The output $I(f)$ contains the intensity and phase information of the RBS. The width of the $|W(f)|$ determines the frequency resolution of OFDR system in frequency domain. Considering that a rectangular window is employed, the $|W(f)|$ is:

$$|W(f)| = \left| \frac{\sin(\pi f T_s)}{\pi f} \right|. \quad (16)$$

The width of $|W(f)|$ is defined by the half width at the first zero point, which is $\Delta f = 1/T_s$. Δf is also the frequency resolution of the output $I(f)$. The spatial resolution of OFDR is then obtained from the **time delay to frequency mapping rate** of $f = \gamma t$ and the velocity of light in fiber:

$$Z_{SR} = \frac{c}{2n} \cdot \frac{\Delta f}{\gamma} = \frac{c}{2n\Delta F}, \quad (17)$$

where $\Delta F = \gamma T_s$ is the frequency sweeping range. According to Eq. (17), the spatial resolution is independent of the sweeping duration, but only determined by the frequency sweeping range. **Therefore, the probe light of OFDR system can have long duration with large sweeping range, and thus the OFDR system can achieve high SNR of RBS and high spatial resolution simultaneously.**

However, due to frequency sweeping nonlinearity and laser phase noise caused by spontaneous radiation, the spatial resolution degrades with the increasing of distance [20]–[22]. There are two ways to eliminate the above mentioned problems. **One way is to employ light source with better performance. Frequency sweeping linearity can be guaranteed with predistortion and feedback control [23], [24], feedforward scheme [25], etc. The other way is the post-processing of the signal. A common method is to sample the photocurrent with equal instantaneous optical frequency interval using an auxiliary interferometer [19], [26]–[29]. Both methods can achieve sub-meter level spatial resolution with tens of kilometer sensing range.**

The main challenge faced by OFDR system when it is adopted for DAS system is the crosstalk problem [10]. **The duration of the probe light is pretty long, and fiber cannot be regarded as stable during one probe when there are vibrations on the FUT.** The probe pulse is phase modulated when it passes by a vibration, resulting in crosstalk from the former vibration to the latter positions of the FUT. This crosstalk gets severe as the frequency of vibration increases. The crosstalk problem limits the application of OFDR system for measurement of dynamic parameters (e.g. vibration). If the crosstalk problem is well solved, OFDR would be a potential method towards high spatial resolution DAS system.

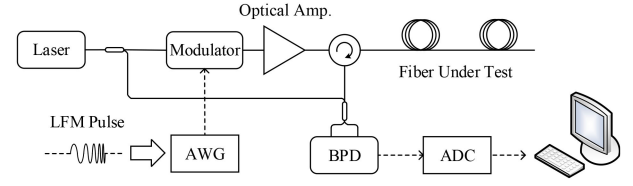


Fig. 5. TGD-OFDR configuration. LFM: linear frequency modulated.

C. Time-Gated Digital Optical Frequency Domain Reflectometry

The time-gated digital OFDR (TGD-OFDR) is a combination of the OTDR and OFDR [11], [12]. System configuration of the TGD-OFDR is illustrated in Fig. 5, which is pretty similar to the configuration of COTDR. The key difference is that the electrical signal generated by the AWG is not a single frequency pulse, but a linear frequency modulated (LFM) pulse. **The pulse width is much longer than that in OTDR system, but shorter than the frequency sweeping duration in OFDR system.**

In TGD-OFDR, the electrical signal generated by the AWG is:

$$s(t) = w\left(\frac{t}{T_p}\right) \exp[j(2\pi f_0 t + \pi \gamma t^2)], \quad (18)$$

where $w(t)$ is the window function, T_p is the pulse width and γ is the frequency sweeping rate. Here rectangular window is applied. Thus, the photoelectrical fields before and after the modulation are:

$$E_l(t) = \sqrt{P_l} \exp[j(2\pi \nu_0 t)] \quad (19)$$

and

$$E_p(t) = \sqrt{P_p} w\left(\frac{t}{T_p}\right) \exp[j(2\pi \nu_0 t + 2\pi f_0 t + \pi \gamma t^2)] \quad (20)$$

respectively.

The photoelectrical field received by the BPD is the summation of all RBS signals along the FUT, as shown in Eq. (21):

$$E_R(t) = \sum_{i=1}^N E_i w\left(\frac{t - \tau_i}{T_p}\right) \exp\{j[2\pi \nu_0(t - \tau_i) + 2\pi f_0(t - \tau_i) + \pi \gamma(t - \tau_i)^2]\}. \quad (21)$$

The photocurrent generated in BPD is:

$$i(t) \propto \Re\{E_R(t) \cdot E_l^*(t)\} = \sum_{i=1}^N A_i w\left(\frac{t - \tau_i}{T_p}\right) \cos \times [2\pi f_0(t - \tau_i) + \pi \gamma(t - \tau_i)^2 - 2\pi \nu_0 \tau_i]. \quad (22)$$

Hilbert transform is performed to the photocurrent to convert the real signal to analytic signal and Eq. (22) can be rewritten as an integral:

$$i(t) = \int_0^{T_F} A(\tau) w\left(\frac{t - \tau}{T_p}\right) \exp \times \{j[2\pi f_0(t - \tau) + \pi \gamma(t - \tau)^2 - 2\pi \nu_0 \tau]\} d\tau = h(t) \otimes s(t), \quad (23)$$

where $h(t)$ is the impulse response of the FUT:

$$h(t) = A(t) \exp(-j2\pi\nu_0 t). \quad (24)$$

According to Eq. (23), the spatial resolution is still determined by the pulse width. To improve the spatial resolution, a matched filter $s^*(-t)$ is generated in digital domain to compress the pulse:

$$I(t) = h(t) \otimes s(t) \otimes s^*(-t) = h(t) \otimes S(t), \quad (25)$$

where $S(t)$ is the compressed pulse:

$$S(t) = W \left(\frac{t}{2T_p} \right) \frac{\sin[\pi\gamma t(T_p - |t|)]}{\pi\gamma t} \times \exp \left\{ -j2\pi \left(f_0 + \frac{\gamma T_p}{2} t \right) \right\}. \quad (26)$$

$S(t)$ is regarded as the effective probe pulse and the spatial resolution of TGD-OFDR is determined by the FWHM of the main lobe of $|S(t)|$ [12]:

$$Z_{SR} = \frac{c}{2n\Delta F}, \quad (27)$$

which is identical to the spatial resolution of the OFDR.

Similar to OFDR, the spatial resolution is determined by the frequency sweeping range, and independent with the probe pulse, so high spatial resolution can be achieved without degeneration of SNR. Moreover, different from OFDR, the probe light in TGD-OFDR has a moderate duration, so the phase noise and crosstalk problem in OFDR are effectively mitigated. Thus, 100-km level sensing range and meter-level spatial resolution are achieved simultaneously without phase compensation [12].

As the analysis above, the reflectometry is a linear time-invariant system. The output is the convolution of the impulse response of the fiber (which contains the vibration signal) and the input probe pulse. They all can be adopted for DAS if the vibration information contained in the RBS signal is correctly demodulated.

III. VIBRATION RECOVERY Via PHASE MODULATION

There are currently two methods to demodulate the vibration information from the RBS. One is to directly calculate the phase of the RBS, and then perform a differential operation to recover the vibration according to the change of phase. The other one is to scan the frequency of the laser to compensate for the RBS intensity change caused by the vibration. Both of the two methods show linearity to the amplitude of the applied strain to fiber. The principle and methods of phase demodulation will be discussed in this section, and the frequency demodulation technologies will be introduced in the next section.

A. Relation Between Strain and RBS Phase

To simplify the analysis, suppose that the fiber is filled with individual reflectors along the fiber. The phase of reflected lightwave is the function of its position z and the applied strain ε . Before the applied strain, the phase of the return lightwave reflected at position z is:

$$\Phi(z, 0) = 2 \int_0^z \beta n(x) dx = 2\beta n z, \quad (28)$$

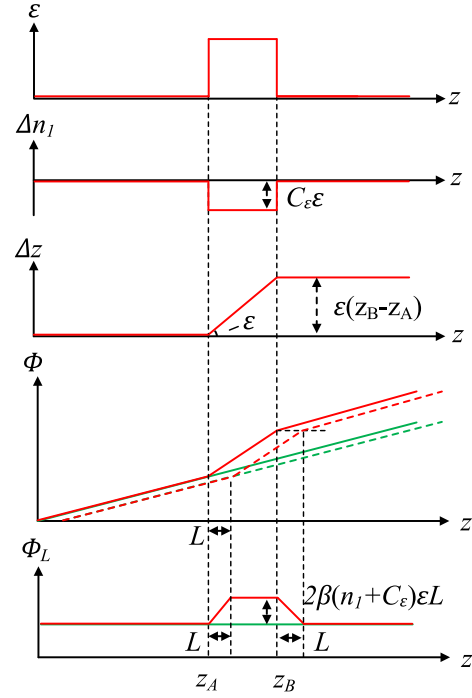


Fig. 6. Principle of RBS phase demodulation. The figures in sequence are: strain distribution along fiber; refractive index of fiber; the optical path variation; the phase of RBS along the fiber (green: no strain; red: strain applied), and phase after spatial differential. L : spatial differential distance.

where Φ is the phase of RBS, $\beta = 2\pi/\lambda$ is the wave vector in vacuum, and the refractive index along the fiber $n(x)$ is a function of strain distribution $\varepsilon(x)$ due to the photo-elastic effect, $n(x) = n + C_\varepsilon \varepsilon(x)$.

Let's consider that a strain ε is applied to a section of the fiber (z_A, z_B), as shown in Fig. 6. For a reflector in $(0, z_A)$, the phase of the reflected light will not change; If the reflector is within the strained section, the phase will be:

$$\begin{aligned} \Phi(z, \varepsilon) &= 2 \int_0^z \beta n(x) dx \\ &= 2 \int_0^{z_A} \beta n(x) dx + 2 \int_{z_A}^z \beta n(x)(1 + \varepsilon) dx \\ &= 2\beta n z_A + 2\beta(n + C_\varepsilon \varepsilon)(1 + \varepsilon)(z - z_A). \end{aligned} \quad (29)$$

If the reflector locates in $z \in (z_B, z_F)$ (z_F is the total length of FUT), the phase of reflected lightwave is:

$$\begin{aligned} \Phi(z, 0) &= 2 \int_0^z \beta n(x) dx \\ &= 2 \int_0^{z_B} \beta n(x) dx + 2 \int_{z_B}^z \beta n(x) dx \\ &= 2\beta n(z - z_B + z_A) \\ &\quad + 2\beta(n + C_\varepsilon \varepsilon)(1 + \varepsilon)(z_B - z_A). \end{aligned} \quad (30)$$

In the section (z_A, z_B), considering two positions with distance of L , their phase difference (distance differential) before

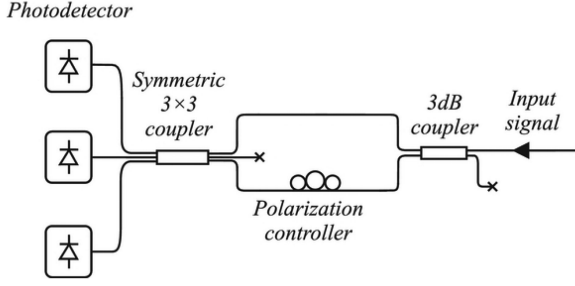


Fig. 7. Scheme using 3x3 coupler [32].

strain is:

$$\Phi_L(0) = \Phi(z, 0) - \Phi(z - L, 0) = 2\beta nL. \quad (31)$$

After the strain, the phase difference becomes:

$$\Phi_L(\varepsilon) = \Phi(z, \varepsilon) - \Phi(z - L, \varepsilon) = 2\beta(n + C_\varepsilon\varepsilon)(1 + \varepsilon)L. \quad (32)$$

So the phase difference before and after the strain (time differential) is:

$$\begin{aligned} \Delta\Phi_L(\varepsilon) &= \Phi_L(\varepsilon) - \Phi_L(0) \\ &= 2\beta(n\varepsilon + C_\varepsilon\varepsilon + C_\varepsilon\varepsilon^2)L. \end{aligned} \quad (33)$$

When the amplitude of the strain is small, the quadratic term can be ignored and above expression is simplified to:

$$\Delta\Phi_L(\varepsilon) = 2\beta(n + C_\varepsilon)\varepsilon L. \quad (34)$$

From Eq. (34), the strain is proportional to the shift of the spatial differential phase, and the strain (vibration waveform) can be obtained linearly. Outside of the vibration area, the phase difference remains 0, which makes it possible to detect the vibration position. When the center wavelength of the laser is 1550.12 nm and the refractive index of the fiber core is 1.46, the sensitivity coefficient in the above formula is calculated to be around 110.37 nε · m/rad [30] [31].

B. Methods of Phase Demodulation

Currently, most DAS systems are phase-demodulation DAS because of the simple structure and small amount of calculation. Several methods have been developed to figure out the phase of the RBS, and great efforts have been made to solve the fading problem.

1) *Unbalanced Interferometer*: One notable technology for RBS phase recovery is by using imbalanced Mach-Zehnder fiber interferometer to introduce a time delay, so the RBS from two positions (spatial differential) could meet at the photo-detector.

A symmetric 3 × 3 coupler could be used at the output of the interferometer [32] to detect the phase, as illustrated in Fig. 7. Compared with the balanced detection, the 3 × 3 coupler and 3 photo-detectors configuration can eliminate signal fading in interferometer. With this method, multiple vibration detection along 1 km fiber with 2 m spatial resolution was realized, with a minimum detectable strain of 80 nε. The 3 × 3 configuration can be replaced by phase generated carrier demodulation [33] or differentiate-and-cross-multiply method [34]. These all are self-interference balance detection.

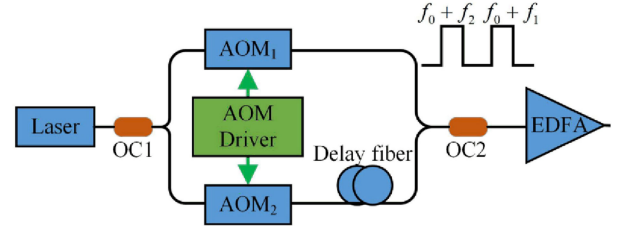


Fig. 8. The pulses injected into fiber in dual-pulse system [37]. OC: optical coupler; EDFA: Erbium-doped fiber amplifier.

2) *Dual Probe Pulses*: Dual-pulse system is another typical direct detection method. As shown in Fig. 8, two pulses with different frequency and proper time delay are injected into the sensing fiber. The RBS of the two probe pulses from different positions will meet at the photo-detector, and the actual waveform is retrieved by heterodyne phase demodulation. In 1990, Dakin and John Philip proposed a double-pulse reflectometer technology [35]. Later A. E. Alekseev *et al.* proposed a DAS system scheme which injects three pairs of probe pulses with different phases [36]. The performance of this structure is further improved by X. He in 2016 [37], in which the response bandwidth ranges from 20 Hz to 25 kHz, and the dynamic range is about 60 dB. One thing should be noticed that in this method, the extracted phase is already the spatial differential phase.

3) *Coherent Detection*: In recent DAS systems, the coherent detection is widely utilized for higher SNR and longer sensing distance. The coherent detection includes heterodyne detection and homodyne detection. In case that the acousto-optical modulator (AOM) is employed to cut the pulse, COTDR based on DAS will be heterodyne detection because of the frequency shift of the AOM (the carrier frequency f_0 in Eq. (2)). Then the phase of the RBS can be retrieved using Hilbert transform. In 2010, Y. Lu and X. Bao introduced the heterodyne coherent detection in their DVS to increase the SNR of RBS [38]; Later, Z. Pan *et al.* proposed a digital phase demodulation algorithm to extract the phase of RBS to realize DAS system [39]. Due to the high SNR and simple phase demodulation algorithm, COTDR based DAS has been extensively studied [40] for further performance improvement.

For homodyne detection, the I/Q demodulation is introduced to extract the phase of RBS [41], [42]. The RBS is split into two beams by the hybrid, and one beam is introduced by 90° phase shift. Then the two beams are mixed with the LO respectively, as illustrated in Fig. 9. Each of the two outputs of the hybrid is detected by a balanced photo-detector, and the AC components of the photo-currents are:

$$\begin{cases} I = \frac{1}{2} \cdot E_s \cdot \cos(2\pi f_0 t + \varphi + \varphi_0) \\ Q = \frac{1}{2} \cdot E_s \cdot \sin(2\pi f_0 t + \varphi + \varphi_0) \end{cases} \quad (35)$$

Where E_s is the amplitude of signal, f_0 is the frequency differentc between signal and LO (it could be zero), and φ_0 is the difference between the initial phases of the LO and the signal. Then the phase of the RBS could be obtained:

$$\varphi + \varphi_0 = \arctan\left(\frac{Q}{I}\right). \quad (36)$$

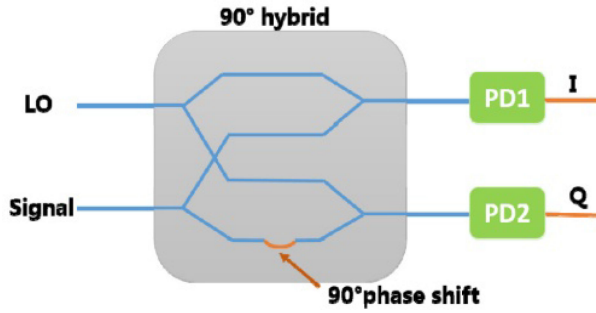


Fig. 9. The homodyne detection with 90° hybrid [41]. LO: local oscillator.

After spatial difference, the phase change caused by vibration could be demodulated, and in [41], the vibration is successfully detected with a sensing range of 12.56 km and a spatial resolution of 10 m. Compared with heterodyne detection, the homodyne detection have smaller receiver bandwidth at the cost of doubled photo-detectors and data acquisition channels.

C. Discussion on Strain Resolution

The strain resolution can assess the sensitivity of DAS, because the DAS actually detect the vibration deduced strain to the fiber. In phase demodulation, the strain resolution is converted from the noise level of the extracted RBS phase, which is further determined by two factors.

1) *The Phase Noise of the Laser Source*: The phase noise of the laser will be directly superimposed on the RBS signal. Numerical simulation indicates that its contribution to the RBS phase noise is roughly proportional to the linewidth of the laser source when the pulse duration is a constant. Meanwhile, the RBS phase noise is proportional to the root of pulse duration when the linewidth of the laser is a constant. So, narrow linewidth laser and short probe pulse will help to suppress the phase noise coupled from the laser source. However, for the typical DAS system based on OTDR configuration with laser linewidth of several kHz and pulse duration of 100 ns (spatial resolution of 10 m), the influence of laser phase noise is found to be negligible compared with the noise induced by insufficient SNR of Rayleigh intensity. On the other hand, the influence of laser phase noise should be considered in DAS system based on TGD-OFDR configuration, and it plays an important role in OFDR configuration.

2) *SNR of the Rayleigh Intensity*: The photo-detector is only sensitive to the intensity of the RBS, and the RBS phase signal is calculated from the RBS intensity signal. So the finite SNR of the Rayleigh intensity brings uncertainty to the extracted phase [43]. Numerical simulation shows that if the intensity noise is white noise, the noise in extracted phase obeys Gaussian distribution as displayed in Fig. 10(a), and the variance is approximately inverse proportional to the SNR of intensity as shown in Fig. 10(b). In the case that the SNR of intensity is greater than 1 (0 dB), the variance of phase extraction noise can be approximately expressed as [44]:

$$\sigma_{\theta}^2 = \frac{1}{\text{SNR}}. \quad (37)$$

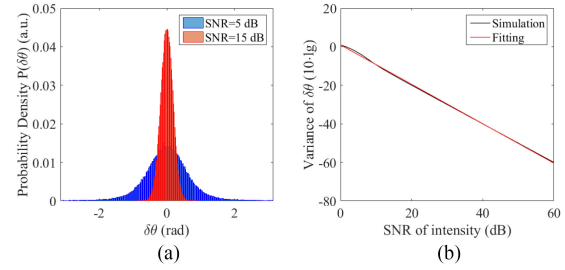


Fig. 10. (a) Probability density function of $\delta\theta$; (b) Relation between the variance of $\delta\theta$ and the SNR of RBS intensity [44].

Obviously a longer probe pulse will bring stronger RBS and higher SNR in average. In OTDR, the high SNR is achieved at the cost of degenerated spatial resolution. The OFDR and TGD-OFDR systems show superiority because they can realize high SNR using long probe pulse and high spatial resolution simultaneously.

It should be pointed out that a long probe pulse does not mean high SNR at all positions along the fiber. Due to the interference within the spatial resolution, the RBS exhibits a strong amplitude fluctuation, and RBS at some random positions will be extremely weak. This phenomenon is well known as the interference fading. In coherent detection, the random polarization state of RBS will result in mismatch with the LO. In this case the signal would decrease too, and this phenomenon is called polarization fading. In those fading points, the poor SNR leads to strong phase fluctuations, so the strain resolution of these places is extremely poor, which is known as dead zone. It is a severe problem in phase demodulation DAS.

D. Solving the Fading Problem

The polarization fading could be solved with polarization diversity reception or polarization maintaining configurations [45], but the interference fading is more difficult to deal with. Many complex algorithms have been proposed to suppress this problem [43], [46]–[49]. The fading position is sensitive to the frequency of probe laser. So the effective solution includes two steps: First is to interrogate the FUT by probes with different frequencies or phases, and the fading positions of different interrogations will not be overlap; the next step is to properly combine the result of each interrogation to produce the vibration signal. In 2012, Z. Pan and H. Cai *et al.* injected two pairs of probe pulses with abrupt phase changes to suppress the interference fading [50], [51]; later J. Zhou *et al.* used probe pulses containing multiple frequency components to eliminate interference fading [52]. Actually the above work focuses on the first step, but the second step is not well solved.

Based on the TGD-OFDR configuration, an inner-pulse frequency-division and a rotated-vector-sum method were proposed in 2017 [53]. In TGD-OFDR, the probe pulse is a wide-band linear frequency swept pulse, so the RBS signal is also linearly swept. The RBS could be divided into three or more different sub-bands by corresponding matched filters. All those sub-bands with different center frequencies could be obtained through single-shot acquisition, which will not decrease the

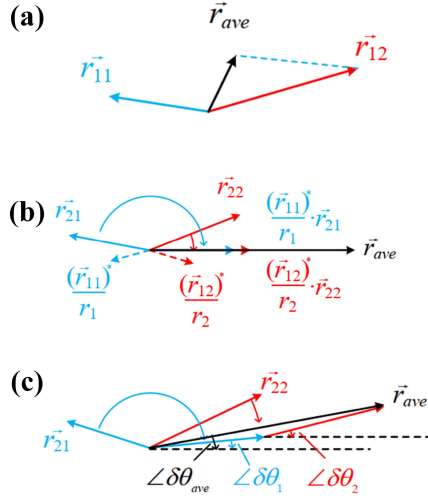


Fig. 11. Schematic of rotated-vector-sum method [53]. (a) Addition of two complex reflection vectors; (b) rotated-vector-sum method without applied strain; (c) rotated-vector-sum method with strain applied.

response frequency of the DAS system. The sub-bands can be regarded as probes with different frequencies. The probability of overlap of fading positions of all the sub-bands is ignorable in practice.

Figure 11 shows how the RBS of different frequencies are summed together before the phase extraction. Before the vibration, the vector $\vec{r}_{11}$ can be rotated to zero-angle by timing its normalized conjugation $\frac{\vec{r}_{11}^*}{|\vec{r}_{11}|}$, as shown in Fig. 11 (a, b). If strain is applied to this section of fiber, the RBS vector $\vec{r}_{12}$ will have angle $\angle\delta\theta_1$ after rotation. The vector of the other sub-band, $\vec{r}_{22}$, is processed in the same way. Then all the rotated vectors are added and its angle $\angle\theta_{ave}$ indicates the phase shift caused by strain. The rotated-vector-sum method is actually a weighted average of the phase variances obtained by different probe pulses, and the weight is the corresponding intensity of RBS. The contribution of phase noise due to weak RBS of one probe pulse is thus minimized. Experimental results show that with three sub-band division, the interference fading can be effectively suppressed over the whole fiber [53].

IV. VIBRATION RECOVERY Via FREQUENCY DEMODULATION

In addition to phase-sensitive demodulation, frequency-sensitive demodulation has also attracted a lot of attention in recent years. The Rayleigh frequency demodulation uses the Rayleigh interference pattern to demodulate the strain distribution quantitatively. It is robust, and immune to coherent fading [54]. Another advantage is that it can detect the static signals, because each strain corresponds to a unique Rayleigh interference pattern [55]. As a comparison, the phase shift of RBS always falls into the range of $-\pi$ to $+\pi$, so the phase demodulation DAS can't map the phase to strain one-by-one especially when the strain is large.

A. Relation Between Strain and Shift of RBS Spectrum

The inhomogeneities of fiber can be modeled as many continuous small scattering points, which can be further regarded

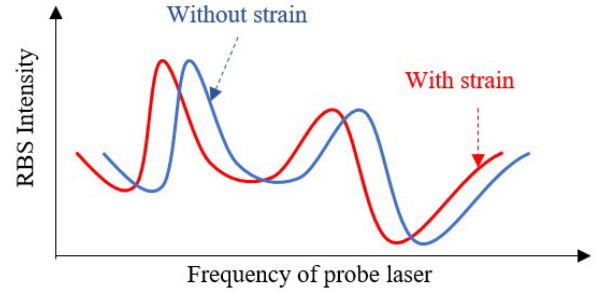


Fig. 12. Principle of frequency demodulated DAS.

as random fiber gratings. The spectrum of each piece of random grating forms the interference pattern. When the random grating experiences strain, as shown in Fig. 12, its spectrum will shift accordingly, similar to the case of fiber Bragg grating (FBG). Different from FBG, the spectrum of random grating do not have a main peak. Instead, the shape of spectrum is chaotic. **So the spectrum shift should be determined by algorithms such as cross-correlation.**

The relations between spectrum shift $\Delta\nu$, refractive index and the strain ε (or temperature ΔT) are described by:

$$\frac{\Delta\nu}{\nu_0} = \frac{\Delta n}{n} = -K_\varepsilon \varepsilon \approx -0.78\varepsilon, \quad (38)$$

where K_ε is the strain coefficient, ν_0 is nominal laser frequency, and n is effective refractive index of fiber. From Eq. (38), the strain is proportional to the shift of the frequency, so the strain (vibration waveform) can be obtained. In standard single mode fiber, the sensitivity coefficient K_ε in Eq. (38) is about 0.78. **When the center wavelength of the laser is 1550 nm, a 1 $\mu\varepsilon$ strain corresponds to a frequency shift of about 150 MHz [56].**

The frequency method needs to perform cross-correlation calculations after obtaining the Rayleigh pattern, and then find the shift of the correlation peak. The resolution of the peak position estimation determines the strain resolution. One factor that limits the strain resolution is the frequency tuning step of the probe laser. In practice, the step of frequency cannot be sufficiently small due to interrogation time and device cost. Interpolation can be used to increase the step resolution in data processing [57]. The other factor is the SNR of RBS, which is consistent with the phase method.

B. Methods of Frequency Demodulation

The frequency demodulation DAS requires the RBS intensity at different frequencies, which makes the system relatively complicated and time-consuming. Besides, detecting the frequency shift of the RBS curve also involves a large amount of calculation. **Several works have been done to solve those problems. At the initial stage the frequency demodulation method can only be used for quasi-stationary signal sensing. Later it is extended to acoustic signal sensing.**

1) *Step-By-Step Frequency Sweeping*: The straightforward method to get the RBS curves at different frequencies is to interrogate the FUT with stepped probe frequency based on OTDR configuration [58], [59]. Then the cross-correlation algorithm is carried out to detect the frequency shift caused by environment

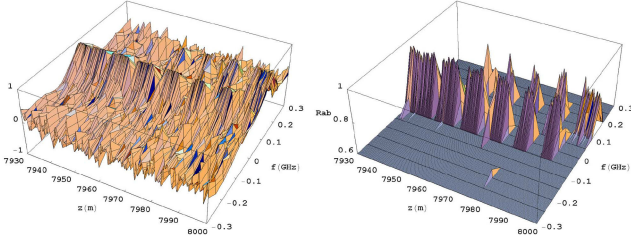


Fig. 13. Cross correlation between vibration and non vibration regions with full scale on vertical axis and extended scale on vertical axis respectively [58].

strain / temperature change, which is a common method that has been extensively studied for time delay estimation.

The cross correlation operation corresponds to a windowed convolution of the RBS curves, and large errors might occur if a strong peak enters the window. So in practice, the Pearson correlation coefficient is used [60], which has better robustness:

$$R_{ab}(f) = \frac{\sum_i [p_a(\nu_i) - \bar{p}_a][p_b(\nu_i + f) - \bar{p}_b]}{\sqrt{\sum_i [p_a(\nu_i) - \bar{p}_a]^2 \sum_i [p_b(\nu_i + f) - \bar{p}_b]^2}}. \quad (39)$$

As depicted in Fig. 13, the peak frequency of cross-correlation indicates the frequency shift. In the work [58], they successfully achieved temperature measurements for an 8-km-long fiber with a resolution of 0.01 °C. However the response bandwidth of system is severely limited by tuning the frequency step by step repeatedly. The measurement time is multiplied by the number of pulse frequency steps because the interval of pulse sequence should be larger than the round-trip time in the fiber. With modified hardware [61] and data processing methods [62], [63], spatial resolution and measuring speed are improved with similar configuration.

2) *Single-Shot Measurement*: Obtaining the RBS intensity at different frequencies step-by-step is time-consuming, which limits the response bandwidth of DAS. So single-shot interrogation with frequency-chirped pulses is proposed, which can significantly accelerate the measurement speed [64].

Similar to the inner-pulse frequency-division in Section III, the wideband RBS is divided into many sub-bands with different center frequency simultaneously by matched filters, so the RBS curves at different frequencies over the whole FUT are obtained. The RBS traces from the first pulse are recorded as the reference, as illustrated in Fig. 14(a), providing the initial state of the FUT. The following probe pulses are processed in the same way to measure the RBS intensity. The highest similarity RBS trace between the reference and following pulses can be obtained by Pearson correlation coefficient, whose frequency shift corresponds to the strain of the fiber. By replacing the step-by-step frequency sweeping with signal-shot pulses, vibration response bandwidth reached 5 kHz over 10 km fiber [64]. In [65], the sensing distance is extended to 75 km with similar method.

3) *Strain-Time Mapping by Chirped Pulse*: The cross correlation in frequency demodulated DAS involves a lot of calculation, which is a big problem in practice. To reduce the amount of calculation, strain-time mapping by chirped pulse method was proposed [66], [67]. When the fiber is strained, its spectrum will shift accordingly. Since the probe is a linearly chirped pulse, the

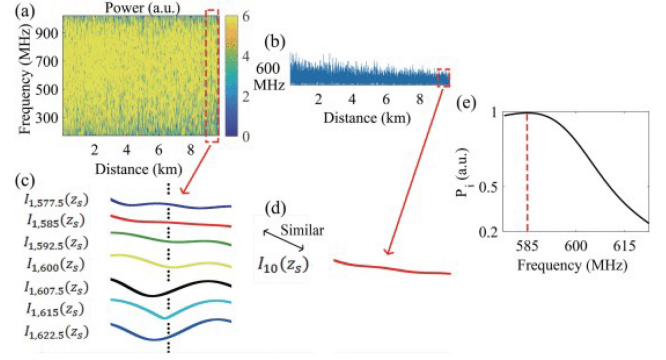


Fig. 14. Demodulation using RBS curve matching [64]. (a) The matrix of RBS traces obtained from the 1-st probe pulse; (b) RBS trace of the 10-th pulse at 600 MHz; (c) Segments of matrix from (a) around [9906, 9906.9] m; (d) The trace segment from (b) around [9906, 9906.9] m; (e) The Pearson correlation coefficient between 1-st pulse (from 577.5 MHz to 622.5 MHz) and the 10-th pulse (600 MHz).

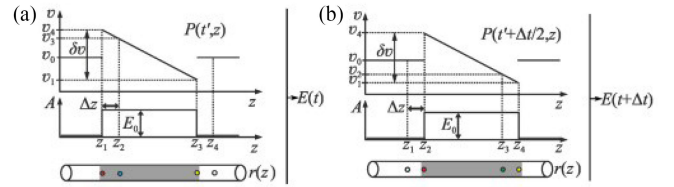


Fig. 15. Reflection of the pulse as it propagates along the fiber [66].

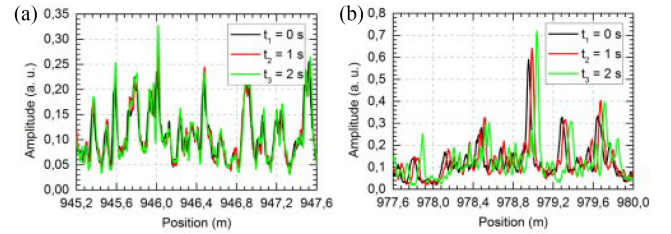


Fig. 16. Longitudinal shift of the RBS trace when temperature changes are applied to the fiber [66].

spectral shift will be equivalent to a time delay of the probe, and thus the RBS curve will be delayed, as illustrated in Fig. 15 and Fig. 16. So the vibration position can be detected by the variance of the RBS intensity, and then cross-correlation will be performed only around the vibration to retrieve the waveform.

The strain-time mapping was first realized in direct detection OTDR configuration. Then it is extended to TGD-OFDR configuration, by which the trade-off between spatial resolution and the length of fiber is removed. A non-matched filter is proposed instead of the matched filter to compress the chirped long duration probe pulse into a short pulse with different chirp rate [68]. Figure 17 shows how the chirped pulse is compressed by matched and non-matched filters. The effective spatial resolution is determined by the compressed pulse, while the pulse energy is the same as the long pulse. A $n\epsilon$ -scale strain signal is detected with a high SNR of 35 dB, and the sensitivity is adjustable in data processing by choosing different non-matched filters.

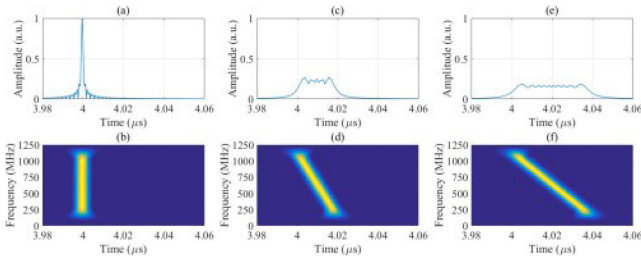


Fig. 17. (a)(c)(e) Chirped pulse and (b)(d)(f) short time Fourier transform of corresponding pulse [68].

V. APPLICATIONS AND DATA PROCESSING TECHNOLOGIES

With the improvement of the technology and the great demand from industry, the DAS has been adopted in many fields. The applications of DAS can be roughly cataloged into two types. In the first type, the DAS can replace the traditional strain sensors or accelerators to form a large scale and low cost sensor array. It should recover the waveform of the mechanical vibration with high fidelity. Examples can be found in the seismic wave detection. The second type of applications are about event detection and classification, such as intrusion detection. The DAS system produces massive data continuously, but only a little mount of data is meaningful. Data processing technologies like treat classification or pattern recognition is necessary. Here some examples of those two types of applications will be discussed.

A. DAS for Vibration Waveform Recovery

Compared with the point-type electronic sensors in traditional vibration detection, DAS has high temporal and spatial accuracy detection capabilities, which can convert optical fibers into large sensor arrays, and realize geophysical exploration with low cost and high accuracy. The DAS has been especially studied in fields of vertical seismic profiling (VSP) for exploration, and the seismic monitoring for geophysical applications.

1) *Das for Vsp*: In VSP technology, many geophones at various depths in a borehole are employed to recorded the seismic waves excited by sources at the surface. The VSP provides a powerful tool to provide the seismic image of the subsurface and to obtain rock properties (velocity, impedance, attenuation, anisotropy) in depth [69]. However, the widespread use of VSP has been hindered by the high cost and risk of deploying electronic geophones in a borehole. The DAS provides a new solution to detect the seismic wave under borehole. Permanently installed fiber cable takes the place of the electronic geophones along the well with lower cost and capability of non-intrusive geophysical monitoring. In 2011, Mestayer *et al.* for the first time demonstrated that DVS was capable of VSP acquisition [70]. The system was based on a COTDR configuration with a sampling rate of 20 kHz. Useful in-well velocity data were gathered from the entire length of the well, which compared well with geophone recording.

However, the DVS was only able to measure seismic intensity. Later, DAS was adopted for seismic measurements because of capability of accurate representation of waveforms, both in terms of amplitude and phase. In [71], a DAS system based on OTDR

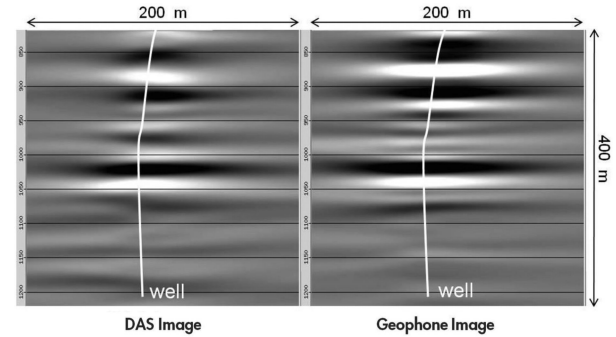


Fig. 18. Walk-away VSP images from conventional geophone data and DAS [71].

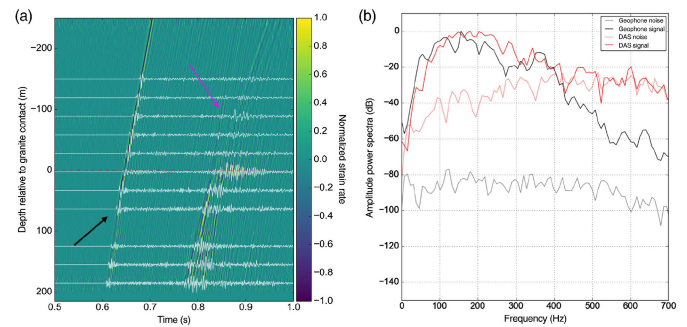


Fig. 19. (a) DAS strain rate (color image) and geophone acceleration records (white lines) of the microseismic event. (b) Signal-to-noise comparison for the geophone marked by black arrow in (a) and the collocated DAS channel. Signal and noise were estimated in 0.5 s windows around the earthquake arrival and the time period before the event, respectively [72].

configuration and dual probe pulses for RBS phase recovery was employed for the VSP measurement. An example of DAS imaging of target area with cable on tubing and under complex overburden is shown in Fig. 18. It is a walk-away image where the imaging target was extremely weak and under complex overburden that excited a lot of multiples. After processing and migration, the DAS image is sufficiently similar to that from geophones in the same well. In [72], a comparison between DAS and geophones indicates that the DAS processing results outperform a regional and local surface array compared with shallow borehole geophones, as shown in Fig. 19.

A quantitative analysis on the quality of DAS in VSP also shows that the DAS data provides reliable amplitude information suitable for creating seismic images [73]. The DAS has several critical advantages: A modern DAS deployment includes much more (thousands or even more) sensing elements than the geophones; The frequency response of DAS is far more broadband than typical inertial geophones with flat response down to 0.008 Hz or lower [74]. Although single DAS channels are substantially noisier than geophones at the same location, their large number and spatial coherency allow for the application of effective array processing techniques. Besides, the DAS holds several other practical benefits like heat resistance, capability of coworking with active well, and the long-term monitoring.

2) *DAS Cables for VSP*: In addition to the interrogation technologies of DAS, which are discussed in the previous sections,

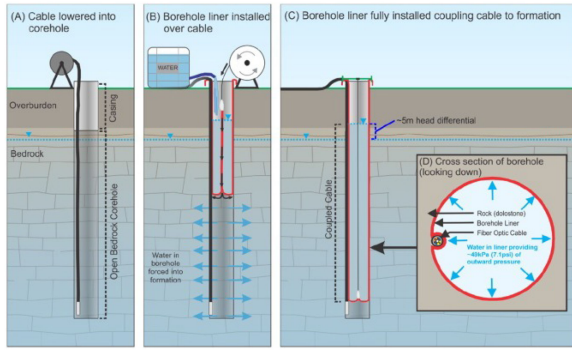


Fig. 20. Schematic showing the installation procedure for the fiber optic cable and borehole liner: (A) Cable lowered into the borehole; (B) borehole liner clamped to top of casing and water is added to evert liner down the borehole; (C) liner is fully installed and tethered off; (D) (inset) cross section looking down the lined borehole demonstrating how the liner forces the cable against the borehole wall [76].

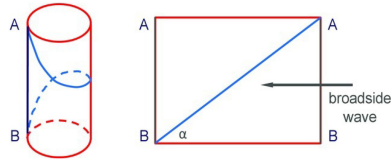


Fig. 21. Helically wrapped fiber for a broadside sensitive cable. Left: fiber (blue) in the cable (red). Right: cable cut along line AB and flattened to obtain a surface in which the wrapping angle α is defined [77].

great efforts have been made on the deployment of fiber cable in the well. A new type of optical fiber (Constellation optical fibre) with enhanced Rayleigh backscattering coefficient is adopted instead of standard single mode fiber to improve the SNR of RBS signal [75]. In a qualification field trial, it has also been demonstrated that the higher backscatter light generated along the Constellation fibre can compensate for additional umbilical and connector losses and provide a practical solution for subsea well installations.

To ensure good cable coupling with the borehole wall, a novel cable coupling solution using a removable and relatively inexpensive FLUTe flexible borehole liner was proposed [76]. Figure 20 shows the installation procedure for the fiber optic cable and borehole liner. The field experiments conducted to assess and validate the technique demonstrated a marked improvement in VSP data quality when with the proposed design. The work showed that tight buffered fibers are more sensitive to dynamic strain than loose tube fibers.

The DAS is only sensitive to axial strain of the fiber, showing a stronger angular dependence than that of a geophone. To overcome this limitation, broadside-sensitive cables have been developed. An example is the helically wrapped cable [77], as shown in Fig. 21. When the wrapping angle α is set to 30° , the helically wrapped cable has a nearly isotropic response. But this cable can't be used to derive the direction of incidence of seismic waves, which still a challenge for the DAS cables.

3) *DAS for Seismic Wave Measurement*: The communication optical fiber cables can be transformed into dense seismic sensor arrays with DAS. Lindsey *et al.* analyzed the seismic observation results of three DAS arrays with different horizontal geometric

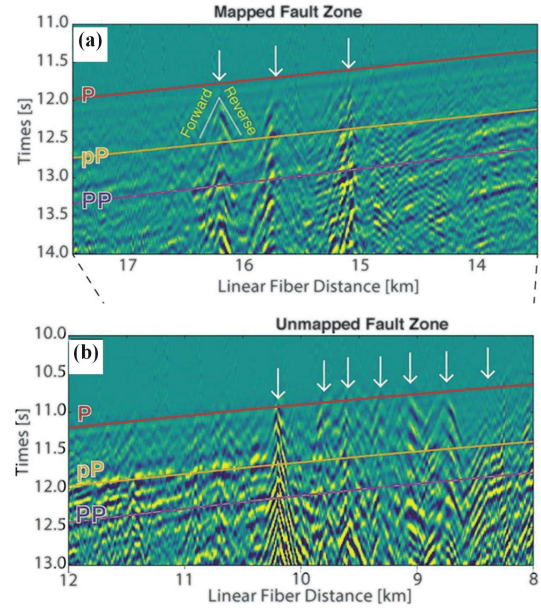


Fig. 22. Earthquake wavefield detection based on existing subsea cables on DAS technique [80]. (a) marked fault zone on map. (b) Unmarked fault zone on map.

shapes, and find the stacking ground motion records shows a high degree of correlation in amplitude and phase with a co-located inertial seismometer record at 0.8–1.6 Hz [78]. In [79], Ajo Franklin *et al.* used DAS deployed on regional unlit fiber in telecommunication cable (so-called dark fiber) for broadband seismic monitoring of both near-surface soil properties and earthquake seismology, and the observation results show that the optical fiber network can be effectively used for many earth science observation tasks. In 2019, Lindsey *et al.* combined DAS with existing dark fiber in subsea cables to observe ocean and solid earth phenomena [80]. Through the detection of the microseism band, multiple submarine fault areas have been discovered, as shown in Fig. 22, and the effects of multiple types of environmental noise such as sub-microseisms, tides, sandstorms, and long gravity waves have been studied and analyzed.

In addition to the monitoring of natural geological activities, ambient-noise can also be recorded by the DAS for monitoring the Earth's near surface. The DAS enables both high sensing channels over a long distance and long-term operations, so time-lapse imaging of shear-wave velocity structures can be realized by combining ambient noise interferometry and multichannel analysis of surface waves. In [81], with a 110-meter-long DAS array deployed perpendicular to a nearby road, multichannel recordings of surface waves via ambient-noise interferometry were acquired. From 502 groups of data the time-lapse repeatability in the model domain was assessed. Later DAS was deployed to record the microseismic signals and ambient noise at Swiss Alpine glacier [82]. The 1 km long fiber optic cable on a glacier surface produced high-quality seismograms related to glacier flow and nearby rock falls. The DAS provided a significant improvement in stick-slip event location and identifies previously unnoticed due to insufficient seismic sensor coverage in the region.

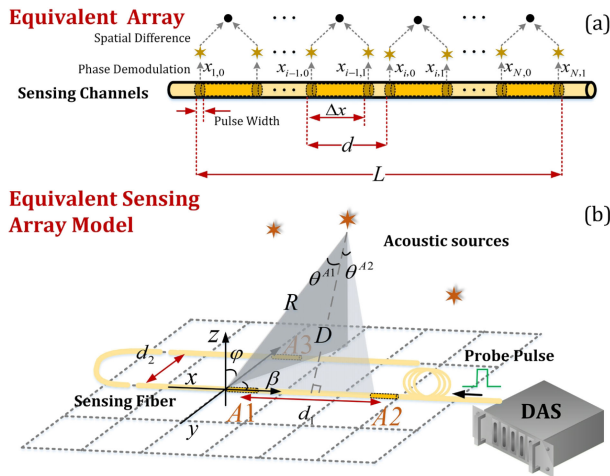


Fig. 23. Model of distributed sensing array and equivalent sensing array [84]. (a) Model of distributed sensing array. (b) equivalent sensing array.

Currently most DAS system can only provide the one-dimensional position of the vibration source along the fiber, without the distance perpendicular to the fiber. In terms of distributed spatial positioning, the authors in [83] detect and localize moving objects in the air using the Doppler effect with the lumped Rayleigh reflectors, which could also be used in water. In [84], Liang *et al.* realized two-dimensional and three-dimensional tracking of acoustic source in air using array signal processing, as illustrated in Fig. 23.

B. DAS for Event Detection

In applications like intrusion detection for perimeter and fence, pipeline surveillance system, railway security, etc., the DAS provides a powerful tool because of the long cover length, linear configuration and high sensitivity. Especially, the deployment of optical fiber cables when building pipelines or railways is a routine well established action, which greatly benefit the adoption of DAS system in practice. In those applications, the actual waveform of vibration is not enough for DAS system. The output data of DAS should be processed in real time and the events of concern should be picked up and assessed within a certain time delay.

The fiber based event detection system are started from DVS. A series of fiber intrusion systems based on DVS system were reported by H. F. Taylor *et al.* [85]–[87], in which a typical phase sensitive OTDR system was proposed. Up to now the DVS is still popular in event detection system, because it is simpler than DAS, and the data processing technologies like pattern recognition can also work without the actual waveform of vibration. However, several drawbacks limit the performance of DVS based system. The harmonic of vibration is inevitable in DVS output because the RBS intensity fluctuation is not proportional to the vibration; The output of DVS is proportional to the power of the light, so its response gets smaller at the far end of the fiber. For those reasons, the DAS becomes the better choice for optical fiber event detection system.

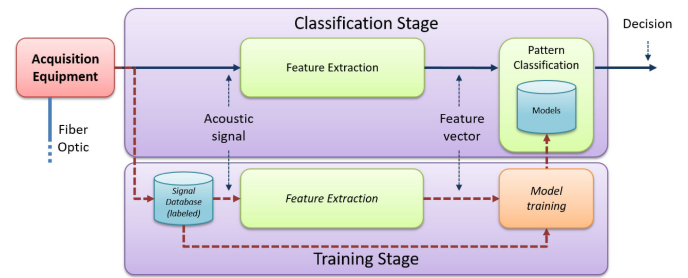


Fig. 24. Standard architecture of a DAS and pattern recognition system [88].

1) *Architecture of DAS Data Processing:* Although the events to detect are different depending on applications, the architectures of DAS data processing is similar. One of the typical applications of DAS is the pipeline surveillance system, which is reviewed in [88]. The pipeline transmission is the most sustainable and safest transmission method to transport oil and gas, and the pipeline integrity is crucial for a safe operation. The DAS technology is specially well suited for the continuous monitoring potential threats to the integrity of pipelines. The information provided by DAS is combined with pattern recognition system and machine learning techniques, so as to provide higher level information and the final decision on the surveilled systems.

An example of DAS data processing system [88] is shown in Fig. 24. The data processing module comprises the Training Stage, which is in charge of generating suitable models that accurately represent the input data characteristics for each of the considered classes, and the Classification Stage, which will generate a final decision according to the input DAS signal and the trained models. In both stages, the signal acquired is given to a feature extraction module, where meaningful patterns (feature vectors) are obtained. Next, the corresponding feature vectors are given to a pattern classification to make a decision, or it is used to generate adequate models in the training stages.

Although the above architecture is for the application of pipeline monitoring, its can also be extended to other applications in which DAS and pattern recognition technology is to be used. The feature extraction and pattern classification are two main steps in the vibration pattern recognition. Some recent progresses on feature extraction and pattern classification technologies will be introduced as follows.

2) *Feature Extraction Technologies:* In the feature extraction step, level crossing rate, which is defined by the rate of crossing through a given threshold in the vibration pulses, is employed in [90]. In [91], the short time Fourier transform (STFT) is used to extract the spectra content, and then the energy of frequency bands is calculated to build the feature vector components. Fourier transform is used to obtain the spectral distribution of the signal in [92], and it is divided into 10 bins to reduce numbers of features. The principal component analysis (PCA) is an effective method to build feature vector, which can reduce the dimension further. In [93], Fourier transform is applied to get the spectrum of each channel after subtracting the average. Then the total energy, the energy of low frequency, the peak value and the

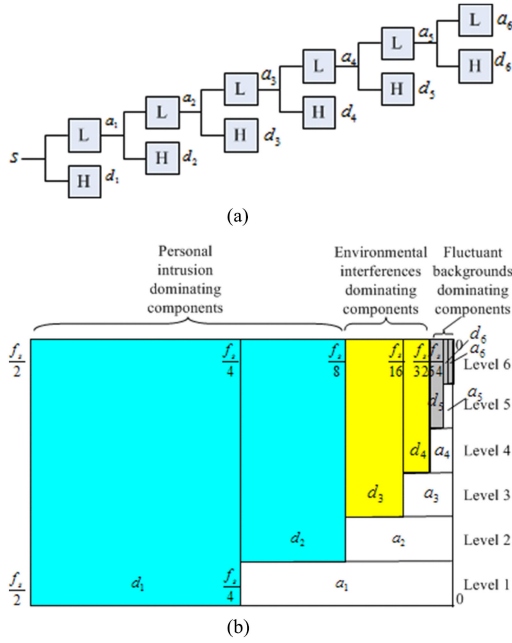


Fig. 25. Scheme of multi-scale wavelet decomposition. (a) Wavelet decomposition based on filter banks. (b) Multi-scale frequency components recombination [89].

mean value of the spectrum are all calculated for each position. Finally, energy ratio of the low frequency to total energy, total energy and ratio of peak value to mean value are selected to form the feature vector.

The multiscale wavelet decomposition method is another common method to extract the features [89]. As illustrated in Fig. 25(a), discrete wavelet transform can be calculated using a set of low-pass and high-pass orthogonal filters, and after passing through the filters, the signal could be decomposed into the approximation coefficients. Different types of events have different frequency components, as shown in Fig. 25(b), so the corresponding frequency components can be recombined on demand. The empirical mode decomposition (EMD) is also adopted in DAS to extract the features [94], which is an adaptive decomposition algorithm and does not need the basis function in advance. In [95], morphological feature extraction (MFE) was employed in DAS. With different types of vibration event, the images in time-space domain are different. At first, segmentation is utilized to separate the event region from the background, and then minimum interval between regions, roundness of the region, pixel number of the region and amplitude of time-space domain signal are selected as the features.

3) *Pattern Classification Technologies*: In the pattern classification step, support vector machine (SVM) is employed as the pattern classification approach in [93]. After feature extraction, feature vectors are put into the SVM, as shown in Fig. 26, and the average recognition accuracy of the 5 types of events was 92.62%. In [92], after Fourier transform and PCA for feature extraction, the SVM is adopted for real-time finding and tracking trains over 15 km. In [96], one-dimensional convolution neural network (CNN) is adopted for feature extraction, and then an SVM is used instead of the softmax layer in the CNN network

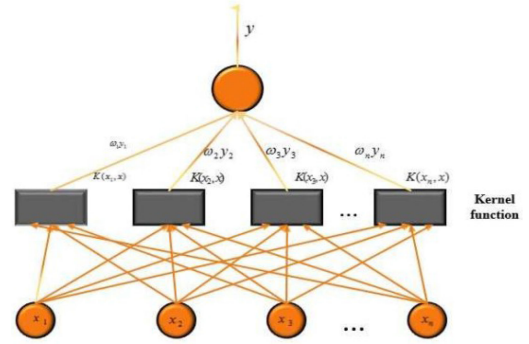


Fig. 26. The model of the SVM [93].

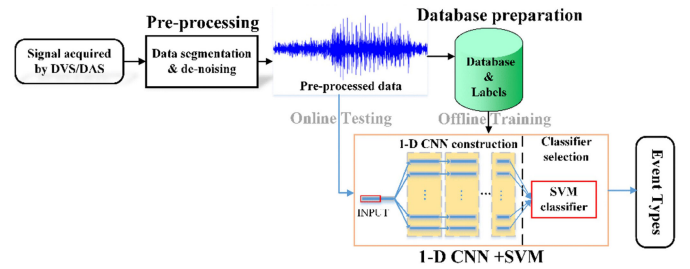


Fig. 27. Construction of the 1-D CNN and SVM [96].

as classifier. Figure 27 shows the construction of the recognition model in the oil pipeline monitoring application.

Relevance Vector Machine technique is used in [95] after feature extraction. Short recognition time and high accuracy are realized. The recognition accuracy can reach 97.87% with 3 events, while the recognition time is within one second. The authors in [89] use 3-layer back propagation artificial neural network to identify the different distributions of three typical events: no intrusion, human intrusion, and hand clapping, and 89.19% identification rate is reached.

In general, the demonstration experiments and reports show that the DAS has achieved great success in certain applications, however, the DAS for event detection system is far from mature or perfect. The main issue is that the pattern recognition algorithms are still case dependent, and there is much room for the improvement of the signal processing technologies.

VI. CONCLUSION AND PERSPECTIVES

In addition to the inherent advantages of distributed fiber sensors such as distributed sensing over long distance, immune to electromagnetic interference, robust against harsh environment, the DAS system has its own characteristics. Since the vibration can propagate a long distance in the medium, DAS can detect vibration events within a certain distance out of the optical cable. With the large communication bandwidth of the optical fiber, the DAS can sample multiple vibrations with high sampling rate, and the amount of information is much larger than the distributed temperature / strain sensors. For those reasons, the DAS is irreplaceable in applications such as seismic measurement, security monitoring of pipelines, power lines, rail track maintenance.

This article reviews the principle of DAS, and introduces the main DAS configurations and technologies. The DAS technology is still developing rapidly. Attentions can be paid to the research fields as follows:

1) Improving performance of the DAS system, such as sensitivity, spatial resolution, dynamic range, frequency response, etc. The RBS is much weaker compared with the fiber gratings or other reflectors, so the sensitivity of current DAS is generally lower than the point sensors by one to two orders; The relatively lower sensitivity is also the main problem in applications like VSP. The spatial resolution is one of the key parameter of DAS. High spatial resolution not only increases the equivalent sensing channels, but also extends the maximal strain/vibration range of the sensor. Another point is the integration with the Brillouin/Raman sensors, realizing multi-sensor information fusion.

2) The DAS signal processing technology, which has two aspects. The first is to make full use of the data from DAS. For example, since the RBS is sensitivity to both strain and temperature, the quasi-static signal of RBS can be used for temperature sensing with high resolution if the cable is properly packaged. The second one is the pattern recognition in event detection, which is the main challenge for many DAS applications. Machine learning technology provides a common framework for DAS signal processing, and has great potential in realization of high performance pattern recognition.

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Zuyuan He (Senior Member, IEEE) received the B.S. and M.S. degrees in electronic engineering from Shanghai Jiao Tong University, Shanghai, China, in 1984 and 1987, respectively, and the Ph.D. degree in photonics from the University of Tokyo, Tokyo, Japan, in 1999.

He was a Research Associate with the Nanjing University of Science and Technology, Nanjing, China, in 1987, where he became a Lecturer in 1990. From 1995 to 1996, he was a Research Fellow studying optical information processing with the Research Center for Advanced Science and Technology, University of Tokyo. In 1999, he was a Research Associate with the University of Tokyo, where he worked on the measurement and characterization of fiber optic components and systems, fiber optic reflectometry, fiber optic sensors, and multidimensional optical information processing. In 2001, he was a Lead Engineer with CIENA Corporation, Linthicum, MD, USA, heading the optical testing and optical process development group. He was a Lecturer with the University of Tokyo in 2003, where became an Associate Professor in 2005 and a Full Professor in 2010. He is currently a Chair Professor and the Head of the Department of Electronic Engineering, Shanghai Jiao Tong University. He has coauthored about 400 papers in peer-refereed journals and international conferences. His current research interests include optical fiber sensors, specialty optical fibers, and optical interconnects.

He is a Senior Member of OSA. Since 2013, he has been an Associate Editor for the IEEE/OSA JOURNAL OF LIGHTWAVE TECHNOLOGY. He was a Technical Program Committee Member in a variety of international conferences, such as the CLEO, OFC, and OFS and the General Chair of Asia Communications and Photonics Conference 2014 and Asia-Pacific Optical Sensors Conference 2016.

Qingwen Liu (Member, IEEE) received the B.S. and M.S. degrees in optoelectronics from Tianjin University, Tianjin, China, in 2005 and 2007, respectively, and the Ph.D. degree in electronic engineering from The University of Tokyo, Tokyo, Japan, in 2012. In 2013, he joined the Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, where he is currently an Associate Professor. His research interests include high resolution fiber grating sensors, distributed optical fiber sensors, and photonic sensors.